

PrepGenie — AI Resume Analyzer & Mock Interview Coach

Executive Project Summary

The Chosen Project

PrepGenie is an AI-powered SaaS platform that helps students and job seekers:

- 1. Get instant, ATS-style resume analysis and improvement suggestions.
- 2. Practice AI-driven mock interviews (technical + HR) with real-time feedback.
- 3. Track their preparation progress on a personal dashboard.

Why This Project (Strategic Rationale)

Criterion	Fit
AI-powered	Uses LLM (via API) for resume parsing, scoring, question generation, and answer evaluation
Deployable	Frontend on Vercel, backend on Render/Railway, DB on Supabase/Postgres — all free-tier friendly
Solves a real problem	Every placement-season student needs resume feedback + interview practice; recruiters value candidates who can talk about "why" they built it
GitHub/LinkedIn impressive	Full-stack, AI integration, auth, dashboards, PDF parsing, real-time chat UI — checks every recruiter box
10-day feasible	Modular scope: Day 2-4 core resume analyzer, Day 5-7 mock interview engine, Day 8-9 dashboard + polish, Day 10 deploy + docs
Internship impact	Demonstrates full-stack + AI + product thinking — exactly what interviewers ask about

Tech Stack (Final Decision — No Further Choices Needed)

- **Frontend:** React (Vite) + TailwindCSS + shadcn/ui + Framer Motion
- **Backend:** Node.js + Express (or FastAPI if student prefers Python — default: Node.js for JS-consistency)
- **Database:** PostgreSQL via Supabase (also gives free auth)
- **AI Layer:** Anthropic Claude API (claude-sonnet-5) for resume analysis, question generation, answer scoring
- **File Parsing:** pdf-parse / mammoth for resume text extraction
- **Auth:** Supabase Auth (email + Google OAuth)
- **Deployment:** Frontend → Vercel, Backend → Render, DB → Supabase
- **Version Control:** GitHub with proper README, LICENSE, CONTRIBUTING.md

Core Modules

1. **Resume Analyzer** — Upload PDF/DOCX → AI extracts skills, experience, scores ATS-friendliness, gives section-by-section improvement suggestions.
2. **Mock Interview Coach** — Choose role (e.g., "SDE Intern", "Data Analyst") → AI asks questions one at a time → user answers (text or voice-to-text) → AI scores clarity, correctness, confidence, and gives improvement tips.
3. **Progress Dashboard** — Tracks resume score history, interview scores over time, weak topic areas, and a study streak tracker.

10-Day Timeline Overview

Day	Focus
1	(Already covered — kickoff/setup context)
2	Project scaffolding, auth, DB schema
3	Resume upload + parsing pipeline
4	AI resume analysis engine + UI
5	Mock interview question generation engine
6	Interview chat UI + answer submission flow
7	AI answer scoring + feedback engine
8	Dashboard: analytics, charts, history
9	UI polish, error handling, edge cases, responsive design
10	Testing, deployment, README, demo video, LinkedIn post prep

Deliverables at the End of Day 10

- Live deployed app (public URL)
- Public GitHub repo with clean commit history
- Full README with architecture diagram
- Demo video script + screenshots
- LinkedIn post draft (optional bonus)
- Resume bullet points describing the project (bonus - ask if needed)

Success Metrics

- Resume analyzer gives structured, actionable feedback in <10s
- Mock interview supports at least 3 role tracks (SDE, Data Analyst, Core/Non-CS)
- Dashboard shows at least 2 chart types (progress line chart, skill radar chart)
- App is fully responsive and has zero console errors on deploy

